

HS-402 ENGINEERING ECONOMICS

ASSIGNMENT#2 SUPPLY DEMAND AND ANALYSIS UNDER ECONOMIC INSTABILITY



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ASSIGNMENT#2

SUPPLY DEMAND AND ANALYSIS UNDER ECONOMIC INSTABILITY

Conduct a thorough study and analyze your product's supply and demand in your current market under the light of the economic instability in the country.

INTRODUCTION

The development of technological products' supply and demand is highly influenced by the global and local economic conditions. In Pakistan, recent economic instability showcased with inflation, currency devaluation, energy tariff hikes, and political uncertainty has significantly affected both producer and consumer behavior. This report contains a detailed study and analysis of supply and demand for an **IOT-based Air Quality Monitoring System (AQMS)**, designed for **environmental monitoring using sensors, microcontrollers, and cloud connectivity**. The analysis will focus primarily on the challenges and opportunities based on the current economy of Pakistan.

MARKET OVERVIEW

Increased concerns air about pollution that is going on within **Pakistan. Lahore, Karachi, and Islamabad are among the afflicted urban centers. IQAir's 2023 World Air Quality Report revealed that Lahore has ranked in the top five most polluted cities worldwide.** When awareness is increased regarding the benefits of environmental health, the demand for affordable, real-time air quality monitoring is also increased. Such types of coverage gaps have been created by the expensive government monitoring stations and less in number. **IOT based AQMS devices are highly cost feasible and scalable, which fit extremely well in households, school setups, small businesses, and local governments.**

DEMAND ANALYSIS

1. DETERMINANTS OF DEMAND:

Due to **inflation exceeding 25% in 2024**, demand for tech products in Pakistan has become quite elastic owing to the decrease in purchasing power of consumers. There has been **greater awareness about environmental issues**, which has kept demand stable with environment-conscious consumers and institutions. **Discounts offered on service utility in decision-making e.g. closing down schools on account of smog enhance the value proposition of the product.** Demand increases during winter smog in Punjab due to pollution levels.

2. CURRENT DEMAND CONDITIONS:

There is secret demand, people want the product but postpone their purchases or look for cheaper options. **Interest among schools, clinics, and municipalities is high, but budget approvals are slow.** We expect a **15-20% decrease in short-term demand due to inflation**, however, we have a **positive outlook for long-term demand as health and safety are increasingly becoming a priority.**

SUPPLY ANALYSIS PRODUCTION INPUTS

Costs of **microcontrollers and sensors like ESP32, PM2.5 sensors and MQ series gas detectors** have gone up owing to import restrictions, currency depreciation and delay of shipping. Hiring of final assembly and testing is now done locally to minimize costs and benefit from local labor since cloud services are **charged in USD as with Firebase and AWS IOT.**

SUPPLY CHALLENGES

Government-imposed import restrictions and tariffs have delayed the acquisition of key sensors and microcontrollers. Higher electricity and transport tariffs have increased **local production costs by 12-18% in the last fiscal year**. Additionally, high interest rates pose challenges for small tech manufacturers in accessing capital to maintain inventory and scale production.

PRICE-QUALITY EQUILIBRIUM SHIFT

The product, **initially priced at PKR 12,000 with a 200-unit monthly demand**, has now been **priced at PKR 15,000 due to inflation and currency issues**, indicating a contraction in supply and demand, **with a decrease in demand to around 130 units**.

STRATEGIC IMPLICATIONS AND ADAPTATIONS

The company has made strategic changes to maintain market relevance and avoid inventory buildup, including introducing a basic version of the AQMS without cloud storage, partnering with **local PCB manufacturers and enclosures suppliers**, offering **flexible payment options to educational institutions and small clinics**, and expanding into online marketplaces like Daraz and Telemart.

FUTURE OUTLOOK

Despite the economic downturn, long-term demand for environmental IOT solutions is promising due to government interest in smart cities and sustainable development goals. Corporate environmental responsibility in industrial sectors could also contribute. Future supply growth depends on macroeconomic stabilization, exchange rate control, inflation reduction, and liberalized import policies.

CONCLUSION

In conclusion, the IOT-based Air Quality Monitoring System in Pakistan faces significant stress due to economic instability, with supply constrained by rising costs and import restrictions, and demand shrinking due to inflation and declining incomes.

References

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